

Konstruksi Perangkat Lunak

PERTEMUAN 2

IDE, GUI Builder, dan Version Control



Ekspektasi Luaran

Mampu memilih dan menggunakan IDE, GUI Builders, dan version control yang cocok dengan kebutuhan yang sudah terspesifikasi.



Outline

- Integrated Development Environment
 - GUI Builder
 - Version Control
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IDE, GUI Builder, dan Version Control

Integrated Development Environment



Why We Need IDE?

40%

of programmer's time is used for editing code

What IDE Offers

- Compilation and error detection from within the editor.
 - Integration with source-code control, build, test, and debugging tools.
 - Compressed or outline views of programs (class names only or logical structures without the contents, also known as “folding”)
 - Jump to definitions of classes, routines, and variables
 - Jump to all places where a class, routine, or variable is used
 - Language-specific formatting
 - Interactive help for the language being edited
 - Brace (*begin-end*) matching
 - Templates for common language constructs (the editor completing the structure of a *for* loop after the programmer types *for*, for example)
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What IDE Offers

- Smart indenting (including easily changing the indentation of a block of statements when logic changes)
 - Automated code transforms or refactorings
 - Macros programmable in a familiar programming language
 - Listing of search strings so that commonly used strings don't need to be retyped
 - Regular expressions in search-and-replace
 - Search-and-replace across a group of files
 - Editing multiple files simultaneously
 - Side-by-side diff comparisons
 - Multilevel undo
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How to Pick the Right IDE?



COST



LANGUAGE



FEATURE

Options of IDE

IDE	Language Supported	Site
Visual Studio	C#, JavaScript, Python, Visual Basic, C++, F#	https://visualstudio.microsoft.com/
Eclipse	Java (Default), Ada, ABAP, C, C++, C#, Clojure, COBOL, D, Erlang, Fortran, Groovy, Haskell, JavaScript, Julia,[7] Lasso, Lua, NATURAL, Perl, PHP, Prolog, Python, R, Ruby (including Ruby on Rails framework), Rust, Scala, and Scheme (plugins)	https://www.eclipse.org/ide/
NetBeans	Java, HTML, JavaScript, CSS, PHP, C/C++	https://netbeans.org/
JetBrains	C++, C, Objective-C/C++, Python, Rust, Swift, Fortran	https://www.jetbrains.com/
Code::Blocks	C, C++, Fortran	https://codeblocks.org

and many more!

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GUI Builder

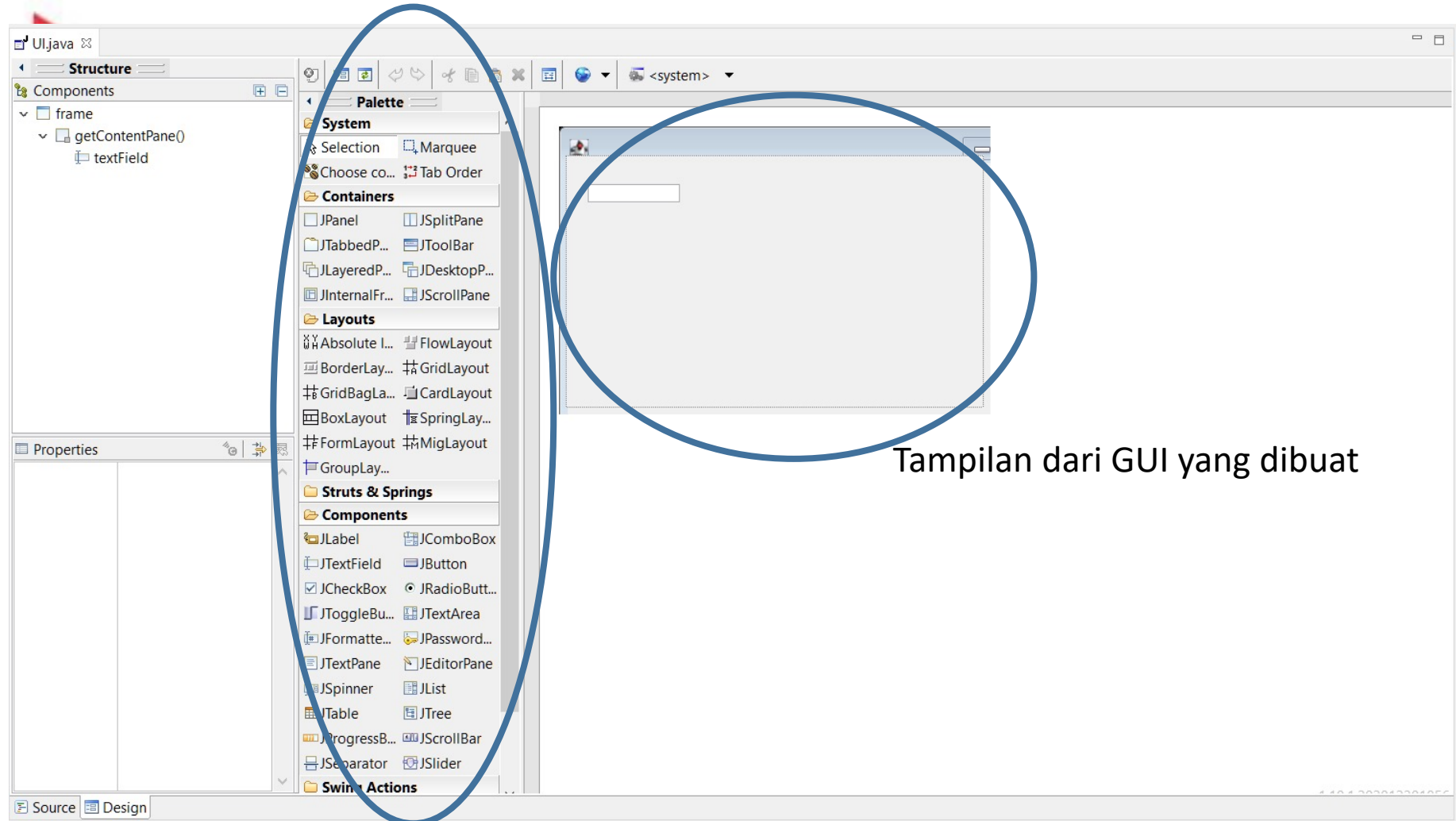


What is GUI Builder?

“A GUI (Graphical User Interface) builder is a software development tool that **enables the developer to create and maintain GUIs in a WYSIWYG (what you see is what you get) mode.**”



Contoh GUI Builder



Tampilan dari GUI yang dibuat

Elemen-elemen GUI

Options of GUI Builder

Language Supported	GUI Builder
Visual C#	Windows Presentation Foundation
C++	Qt
Objective-C/Swift	GNUstep
Java	Android Studio / WindowsBuilder

and many more!



IDE, GUI Builder, dan Version Control

Version Control



Why Version Control?

“A software project is dynamic. The code changes, the design changes, and the requirements change.”



Benefit of Version Control

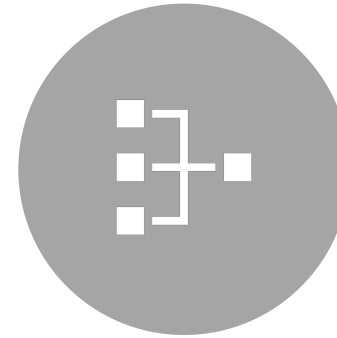
- You don't step on anyone's toes by working on a file while someone else is working on it (or at least you'll know about it if you do).
 - You can easily update your copies of all the project's files to the current versions, usually by issuing a single command.
 - You can backtrack to any version of any file that was ever checked into version control.
 - You can get a list of the changes made to any version of any file.
 - You don't have to worry about personal backups because the version-control copy is a safety net.
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Two Types of Version Control System



CENTRALIZED

(+) More access control
(-) Need dedicated server
Ex: SVN, Perforce



DISTRIBUTED

(+) More reliable
(-) Limited access control
Ex: GitHub, GitLab

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Final Takes



Always Remember!

It is essential to have good tools, but it is also essential that
the tools should be used in the right way.

— *Wallace D. Wattles* —

AZ QUOTES



Q&A

Any questions or comments?



Further reading

- S. McConnell, Code Complete 2nd Edition, Microsoft Press, 1993. Chapter 28.2 and 30.
 - SWEBOK V3.0
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